forestspot

a geospatial application by foresters for foresters

*Making point buffering and pixel extractions from Google Earth Engine datasets **faster** and **easier***

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Rationale

Motivation

- Geospatial data can provide information like environmental variables or remotely sensed vegetation indices
- **But!** Accessing and handling geospatial data can be hard![1]

Solution

- Create a point-and-click oriented application to reduce steps
- Three steps needed to pull pixel values from Google Earth Engine Images and ImageCollections

Usage

Two functionality groups:

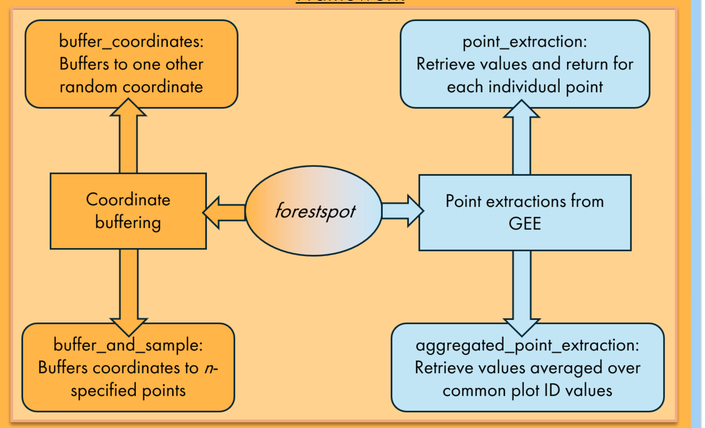
Point extractions

- Performs pixel extractions for provided list of coordinates[2]
- ***Inputs:*** coordinates file (dd.) + select GEE dataset

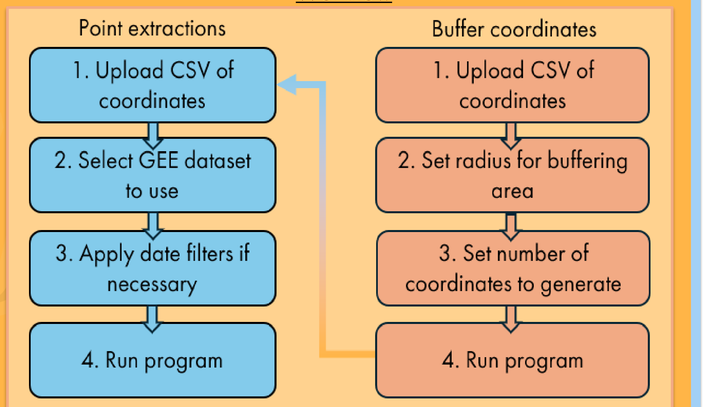
Buffering approaches

- Optional step to buffer sensitive coordinates prior to 'Point extractions'
- ***Inputs:*** coordinates file (dd.), radius *r*, *no. coordinates* to generate

Framework



Workflow

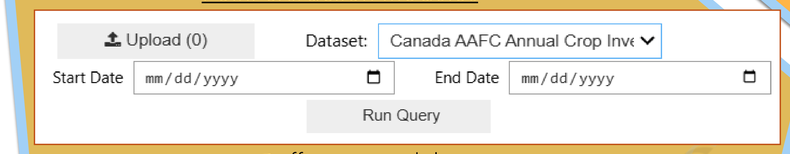


Installation

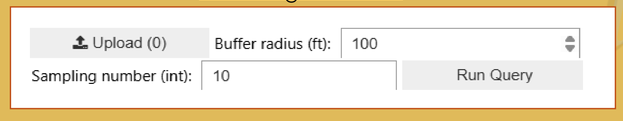
In a Python session:

```
pip install forestspot
```

Point extraction modules



Buffering modules

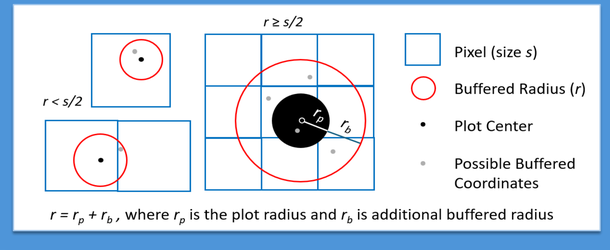


No Python? No problem!

Web-based app

Buffering considerations

How should *radius* and *no. coordinates* be chosen?
It depends ☺



Case study

Scan to view WorldClim dataset info

To demonstrate how *forestspot* can complement existing inventories, we queried environmental data from WorldClim BIO var V1 using the U.S. Forest Service Forest Inventory and Analysis (FIA) inventory plots located on 10 national parks and forests in southern Appalachia[3].

Private FIA plot coordinates' returned values were compared against public FIA coordinates', a single private-buffered coordinate's (r = 1 mi), and 10 private-buffered coordinates' (r = 1 mi) returned values. (WorldClim resolution: 927.67 m)

Percent Differences (%)

BIO var	Public vs. Private (baseline)	Single Buffer on Private vs. Private	10-Point Buffer on Private vs. Private
01	0.45	0.43	0.48
02	0.18	0.19	0.21
03	0.08	0.09	0.11
04	0.03	0.04	0.03
05	0.21	0.21	0.23
...	...	...	...
15	0.34	0.45	0.48
16	-0.14	-0.14	-0.14
17	-0.17	-0.20	-0.18
18	-0.15	-0.15	-0.15
19	-0.14	-0.17	-0.15

Results show that low error exists among all buffering methods compared to raw queried values.

In this case, the public FIA coordinates are sufficient for WorldClim BIO variables.

Citations

[1] Fassnacht FE, White JC, Wulder MA et al. Remote sensing in forestry: current challenges, considerations and directions. Forestry: An International Journal of Forest Research 2024;97(1):11–37. https://doi.org/10.1093/forestry/cpad024.
[2] Wu Q. geemap: A Python package for interactive mapping with Google Earth Engine. Journal of Open Source Software 2020;5(51):2305. https://doi.org/10.21105/joss.02305.
[3] https://developers.google.com/earth-engine/datasets/catalog/WORLDCLIM_V1_BIO

github.com/taraskiba/forestspot
taraskiba.github.io/forestspot/examples/usage

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